

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Science in Information Technology (MSc. (I.T.))

Semester-III University Examination November-2019

MS315 Basics of Cloud Computing

Date: 11.11.19, Monday Time: 1.30 p.m. to 4.30 p.m. Maximum Marks: 70

Instructions:

- 1. Attempt both sections in separate answer books.**
- 2. Figure to right indicates marks.**
- 3. Make suitable assumption when necessary.**

SECTION – 1

- Q.1** Answer following questions: [11]
- i) Define: Cloud Computing
 - ii) What is agility in cloud computing?
 - iii) List three forms of resource pooling architecture.
 - iv) Write the role of automated scaling listener in dynamic scalability architecture.
 - v) List any two essential components of cloud contract.
 - vi) What do you mean by elasticity in cloud computing?
 - vii) Mention any two advantages of hybrid cloud.
 - viii) What do you mean by burst-in for cloud bursting architecture?
 - ix) What is utility computing?
 - x) State any two disadvantages of private cloud.
 - xi) List any two horizontally scaled cloud architectures.
- Q.2 (A)** Write note on Software as a Service (SaaS). [06]
(B) Write note on Public Cloud. [06]
- OR**
- Q.2 (A)** Write note on Platform as a Service (PaaS). [06]
(B) Write note on Community Cloud. [06]
- Q.3 (A)** Explain resource pooling architecture in detail with suitable diagram. [06]
(B) What is pay-per-use model? Draw and explain timeline of cloud computing. [06]
- OR**
- Q.3 (A)** State any four characteristics of cloud computing and explain any two of them. Write need of cloud computing over traditional server concept. [06]
(B) Explain workload distribution architecture in detail with suitable diagram. [06]

SECTION – 2

- Q.4** Answer following questions: [11]
- i) Define: Virtual firewall
 - ii) What do you mean by resource replication?

- iii) What is asymmetric encryption?
- iv) List any four application level security attacks.
- v) What is hashing?
- vi) List any four crucial threats when data are stored in cloud.
- vii) What is virtual network?
- viii) Write full forms of: NFS, SLA
- ix) Enlist any two existing trust mechanisms in cloud.
- x) Which three are biggest cost centers related to the cloud environment?
- xi) List any two hidden aspects of cloud implementation.

- Q.5 (A)** Write note on logical network perimeter in cloud infrastructure mechanism. **[06]**
(B) Write note on digital signature and public key infrastructure. **[06]**

OR

- Q.5 (A)** Write note on resource management system. **[06]**
(B) Write note on cross-site scripting and cookie poisoning. **[06]**

- Q.6 (A)** Explain how security, existing references and interoperability affects the cloud contracts. **[06]**

- (B)** Write note on billing management system for cloud computing. **[06]**

OR

- Q.6 (A)** Why understanding exit policy, and understanding roles and responsibilities are considered as key aspects in cloud contract formulation? Justify. **[06]**

- (B)** Write note on SLA management system for cloud computing. **[06]**